Pijoan-Mas (2006): Precautionary Savings or Working Longer Hours?

June 6, 2026

Model

This note reports a replication of the incomplete-markets economy in [Pijoan-Mas, 2006]. Households solve

$$V(a,\varepsilon) = \max_{c,a',h} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \lambda \frac{(1-h)^{1-\nu}}{1-\nu} + \beta \sum_{\varepsilon'} \Gamma_{\varepsilon,\varepsilon'} V(a',\varepsilon') \right\}$$

subject to

$$c + a' = w\varepsilon h + (1+r)a,$$

and

$$c \geq 0, \quad 0 \leq h \leq 1, \quad a' \geq 0.$$

The individual state variables are asset holdings a and the idiosyncratic productivity shock ε, denoted z in the MATLAB code. The matrix $\Gamma_{\varepsilon,\varepsilon'}$ gives transition probabilities for productivity, with $\sum_{\varepsilon'} \Gamma_{\varepsilon,\varepsilon'} = 1$ for each ε.

Households choose consumption c, hours h, and next-period assets a'. The associated policy functions are denoted by $c = g_c(a,\varepsilon)$, $h = g_h(a,\varepsilon)$, and $a' = g_a(a,\varepsilon)$.

Factor prices satisfy the representative firm's first-order conditions:

$$r = (1-\theta)\left(\frac{K}{L}\right)^{-\theta} - \delta,$$
$$w = \theta \left(\frac{K}{L}\right)^{1-\theta}.$$

The MATLAB code writes the general-equilibrium search in terms of the interest rate r. This is equivalent to searching over K/L, since the firm's first-order condition maps r one-to-one into K/L.

Aggregate output is

$$Y = K^{1-\theta} L^{\theta}.$$

Market clearing requires

$$K = \int g_a(a, \varepsilon)\, d\mu(a, \varepsilon),$$
$$L = \int \varepsilon g_h(a, \varepsilon)\, d\mu(a, \varepsilon),$$

where μ is the stationary distribution. In a stationary equilibrium, aggregate next-period assets equal aggregate current assets, so the code may use either representation for capital clearing. Walras's law then implies the aggregate resource constraint

$$C + \delta K = K^{1-\theta} L^{\theta}.$$

Replication Results

The figures and tables below place the original objects from Pijoan-Mas (2006) next to the corresponding outputs generated by the MATLAB replication.

Table 1
Calibration targets and model parameters

Parameter	Target	Value
σ	$corr(h, \varepsilon) = 0.02$	1.458
ν	$cv(h) = 0.22$	2.833
λ	$H = 0.33$	0.856
β	$K/Y = 3.00$	0.945
θ	$wL/Y = 0.64$	0.640
δ	$I/Y = 0.25$	0.083

Figure 1: Original Table 1 of Pijoan-Mas (2006)

References

[Pijoan-Mas, 2006] Pijoan-Mas, J. (2006). Precautionary savings or working longer hours? *Review of Economic Dynamics*, 9(2):326–352.

Table 1: Calibration targets and model parameters

Parameter	Target	Value
σ	$\text{corr}(h, \varepsilon) = 0.02$	1.458
ν	$cv(h) = 0.20$	2.833
λ	$H = 0.36$	0.856
β	$K/Y = 2.99$	0.945
θ	$wL/Y = 0.64$	0.640
δ	$I/Y = 0.25$	0.083

Table 2
Distributional statistics

Variable	*cv*	Gini	q_1	q_2	q_3	q_4	q_5
Hours							
Model E_0	0.22	0.11	0.21	0.31	0.35	0.37	0.40
Data (CPS)	0.22	0.11	0.24	0.31	0.33	0.35	0.42
Earnings							
Model E_0	0.65	0.33	7.3%	12.4%	17.2%	23.0%	40.1%
Data (CPS)	0.56	0.29	7.9%	13.7%	18.0%	23.3%	37.1%
Data (SCF)	2.65	0.61	−0.2%	4.0%	13.0%	22.9%	60.2%
Wealth							
Model E_0	1.37	0.65	0.1%	2.2%	9.2%	23.1%	65.4%
Data (SCF)	6.53	0.80	−0.3%	1.3%	5.0%	12.2%	81.7%

Notes. *cv* refers to coefficient of variation. $q_1, \ldots, q_5$ refer, for earnings and wealth, to the share held by all people in the corresponding quintile with respect to the total. However, for hours it is the average number of hours worked by people in the corresponding quintile. Statistics from SCF correspond to the 1998 wave and are quoted from Budría et al. (2002). Statistics from SCF correspond to the 2002 wave.

Figure 2: Original Table 2 of Pijoan-Mas (2006)

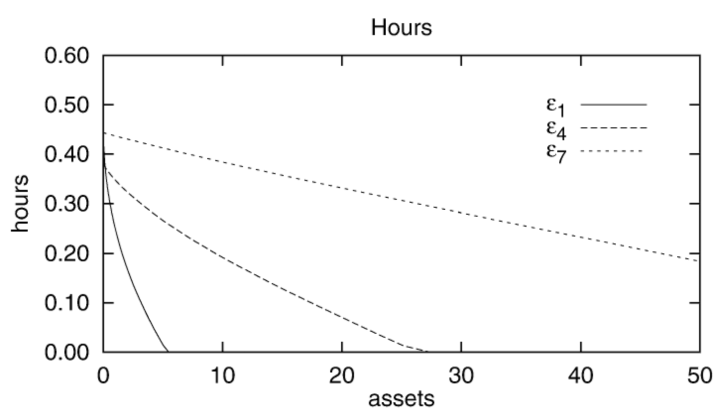

Fig. 1. Policy functions for hours under incomplete markets.

Note. Productivity shocks are labelled such that $\varepsilon_7 > \varepsilon_4 > \varepsilon_1$. Hours are reported as fraction of total available time Assets are reported in levels, where 1.15 corresponds to the (per capita) output of the economy.

Figure 3: Original Figure 1 of Pijoan-Mas (2006)

Table 2: Distributional statistics

Variable	cv	Gini	q_1	q_2	q_3	q_4	q_5
Hours							
Model E_0	0.20	0.10	0.24	0.34	0.37	0.40	0.43
Data (CPS)	0.22	0.11	0.24	0.31	0.33	0.35	0.42
Earnings							
Model E_0	0.64	0.32	7.4%	12.4%	17.3%	23.0%	39.9%
Data (CPS)	0.56	0.29	7.9%	13.7%	18.0%	23.3%	37.1%
Data (SCF)	2.65	0.61	-0.2%	4.0%	13.0%	22.9%	60.2%
Wealth							
Model E_0	1.36	0.64	0.1%	2.3%	9.4%	23.3%	64.9%
Data (SCF)	6.53	0.80	-0.3%	1.3%	5.0%	12.2%	81.7%

Notes. cv refers to coefficient of variation. $q_1, \ldots, q_5$ refer, for earnings and wealth, to the share held by all people in the corresponding quintile with respect to the total. However, for hours it is the average number of hours worked by people in the corresponding quintile. Statistics from SCF correspond to the 1998 wave and are quoted from Budria et al. (2002). Statistics from CPS correspond to the 2002 wave.

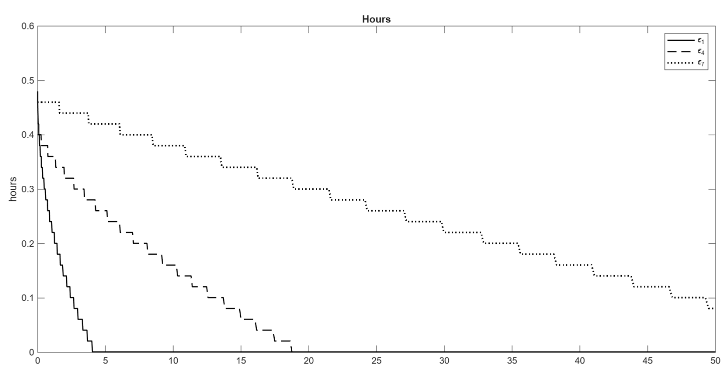

Figure 4: Policy functions for hours under incomplete markets